4.3 Dividing Polynomials

Name _____ Period ____ Date _____

Algebra II 4.3 (30-minute plan) · Student sheet

I can…

- use long division to divide polynomials.
- use synthetic division to divide polynomials by $x - k$.
- use the Remainder Theorem to evaluate polynomial functions.

Part 1 · Notes: fill in the blanks

Long division steps: _____, multiply, _____, bring down.

For synthetic division by $x - k$, put _____ in the box; use _____ for missing terms.

Remainder Theorem: the remainder of $f(x) \div (x - k)$ is _____.

Part 2 · Examples (work them with the class)

Example 1

Divide $(2x^3 + 3x^2 - x + 5) \div (x^2 - x + 1)$.

Example 2

Divide $(x^3 - 3x^2 - 7x + 6) \div (x - 4)$.

Part 3 · Partner practice

Partner practice

Divide $(x^2 + 7x - 2) \div (x - 1)$.